

OSBAMS Pi/Display Power Rev.A - system and Pi power wiring (RC1)

BRANCH 1: OSBAMS Rev.2 controller PCB #1

STM32 / INA228 / relay logic
ISO7721 data isolation to Pi
(PCB #1 ground = 12V_GND)

Waveshare 10.1" DSI display
22-pin DSI to Pi (video/data)
5V/GND to Pi GPIO 5V/GND
(same isolated 5 V domain)

RASPBERRY PI 5 (GPIO header)
5V pins 2 & 4 | GND pins 6 & 9
USB-C power input: NOT connected
PSU_MAX_CURRENT=5000 (after bench checks)

crimped keyed harness
4x 16 AWG, <=150 mm
(no Dupont jumpers)

120 VAC
Mean Well
XDR-75-12
12 V bus

+12 V / GND

BRANCH 2: PCB #2 OSBAMS_Pi_Display_Power_RevA

J_IN
Micro-Fit
430450200

F1
5 A

TVS1
SMBJ15A
C1/C2

U1 RSDW40F-05
9-36 V in -> 5 V / 8 A
1.6 kVDC isolation
ON/OFF, TRIM: unused

F2
8 A

C4 680 uF
C5 10 uF
C6 100 nF
PG LED

J_OUT
430450400
1,2 = +5V_PI
3,4 = PI_GND

isolation barrier

PI_GND is isolated from 12V_GND. Do not connect the isolated 5 V ground to the controller ground.